

Macrolide-resistant *Mycoplasma pneumoniae* resurgence in Chinese children in 2023: a longitudinal, cross-sectional, genomic epidemiology study

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Summary

Background After a prolonged period of low detection rates, *Mycoplasma pneumoniae* resurged in China, during September to November, 2023, raising global concern. This study aims to gain a better understanding of the genetic mechanisms underlying the 2023 increase in cases and the evolutionary dynamics of the epidemic populations, which has been previously hampered due to limited genomic data of this pathogen.

Methods We sequenced 685 *M. pneumoniae* isolates, including 248 isolates from 11 Chinese provinces and municipalities in 2023 and 437 isolates from Beijing (2013–22). By analysing these isolates and 436 publicly global sequences, we reconstructed the pathogen's evolutionary history using time-calibrated phylogenies and effective population size inference. We investigated potential genomic variations contributing to the 2023 resurgence through genome-wide association study and conducted phylogeographic analysis of the 2023 isolates across China.

Findings Two macrolide-resistant epidemic clusters (T1-2-EC1 and T2-2-EC2) were responsible for the 2023 resurgence in China. Both clusters, having acquired the 23S ribosomal RNA A2063G mutation conferring macrolide resistance, emerged in approximately 1997 and 2014, respectively, and subsequently outcompeted their predecessor populations. This coincided with China's large-scale adoption of azithromycin for paediatric community-acquired pneumonia around the early 2000s. Aside from macrolide resistance, T1-2-EC1 independently acquired 17 clade-specific mutations and T2-2-EC2 four clade-specific mutations, which could further explain their increased competitiveness. Whole-genome analysis revealed no resurgence-specific mutations in the 2023 isolates. Phylogeographic analysis showed rapid mixing of T1-2-EC1 isolates between different sampled regions within China.

Interpretation Our study provides evidence that the 2023 resurgence in China is a continuation of the pre-COVID epidemic, rather than emergence of novel variants. The high prevalence of macrolide resistance and rapid intranational spread emphasise the urgent need for enhanced global surveillance of this pathogen.

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Introduction

Mycoplasma pneumoniae is a major cause of paediatric community-acquired pneumonia worldwide.¹ With a genome size of approximately 0.8 Mb, this bacterium possesses a small cell size (1–2 µm long and 0.1–0.2 µm wide) and primarily attaches to the surface of respiratory epithelial cells.² The adhesin P1 protein variation allows classification of *M. pneumoniae* into two major subtypes (subtype 1 and subtype 2).³ Using whole-genome sequencing (WGS) and single-nucleotide polymorphism (SNP) analysis, Hsieh and colleagues⁴ further refined this classification, dividing P1 subtype 1 into clades T1-1, T1-2, and T1-3, and P1 subtype 2 into clades T2-1 and T2-2.

Long-term epidemiological data have revealed that *M. pneumoniae* exhibits cyclical epidemic peaks every 1–3 years.⁵ During the COVID-19 pandemic, the number of reported cases of *M. pneumoniae* dropped significantly worldwide, disrupting this typical pattern.⁶ Following the relaxation of COVID-19 restrictions, the pathogen remained at low levels until a significant resurgence occurred in China, during September to November, 2023.⁷ This delayed re-emergence has been attributed to multiple factors, including the non-pharmaceutical interventions on population immunity and disease dynamics, whereas epidemiological modelling suggests that the relatively long wait between the primary and secondary infection (3 weeks) of *M. pneumoniae* also contributed to this atypical pattern.⁸

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Research in context

Evidence before this study

Following a period of low detection rates during the COVID-19 pandemic, a resurgence of *Mycoplasma pneumoniae* in 2023 raised global attention, particularly in China. On Oct 27, 2024, we searched PubMed for full-text articles published after Jan 1, 2023, using the term “(*Mycoplasma pneumoniae*) AND (resurgence OR outbreak OR epidemic)”. This search returned 185 results. These publications included positivity rate-based epidemic reports that did not incorporate pathogen typing, as well as molecular epidemiology studies incorporating molecular typing, drug susceptibility tests, amplicon sequencing, or a combination thereof. On Oct 27, 2024, we also searched PubMed for full-text articles published after Jan 1, 2023, using the term “(*Mycoplasma pneumoniae*) AND (whole genome sequencing)”. This search returned three results, none of which released genomic sequences of *M pneumoniae* strains isolated after Jan 1, 2023. Two correspondence letters provided preliminary genomic evidence via metagenomic sequencing or whole-genome capture sequencing, one of which reported the 2023 resurgence in China to consist of two epidemic clonal populations, both resistant to macrolide. However, these approaches are limited in their ability to resolve fine-scale genetic diversity, and the restricted temporal and geographical sampling precludes comprehensive analysis of its evolutionary dynamics such as estimated emergence times, variations in effective population sizes, and geographical migration dynamics. Both searches were conducted in published articles of all languages.

Added value of this study

The genomics of the 2023 *M pneumoniae* resurgence remains largely understudied, despite raising considerable attention. Our study addresses this gap by sequencing 685 strains across China (collected in 2013–23, from 11 regions) and integrating them with

436 publicly available sequences. This comprehensive dataset allowed us to investigate the genomic diversity and evolutionary trajectories of *M pneumoniae* in China. Our results estimated that the two epidemic clonal populations emerged in 1997 and 2014, respectively, outcompeted their respective predecessor populations in China during the last decade, and contributed most of the epidemic strains in 2023. Comparing the latest strains isolated in 2023 to strains from earlier years of the same clonal population revealed that no resurgence-associated genomic variation was acquired in either population in 2023. By collecting samples in multiple regions across China in 2023, to our knowledge, this is the first study to show a rapid rate of intranational migration of this pathogen.

Implications of all the available evidence

The findings of this study provide insight into the epidemiology of *M pneumoniae* in China, which serves as guidance for future prevention and control of this pathogen. Our study suggests that the 2023 resurgence in China is a continuation of previous epidemic waves, rather than the emergence of novel strains. The timeframe of one epidemic population outcompeting its predecessor coincides with China's initiation of the large-scale application of azithromycin in the treatment of paediatric community-acquired pneumonia. Beyond macrolide resistance, both epidemic populations acquired distinct clade-specific mutations that might have contributed to their epidemiological successes. These mutations are promising targets for future functional studies. The rapid spread of this pathogen in China poses a significant public health risk, emphasising the need for global surveillance and sequencing efforts to prevent the potential global spread of these drug-resistant strains in the future.

Li and colleagues⁹ analysed Chinese *M pneumoniae* strains from 2023 through metagenomic sequencing, combined with previously published isolate data. Their findings revealed that the 2023 strains belonged to two epidemic clusters: EC1 (within the T1-2 clade) and EC2 (within the T2-2 clade).⁹ However, since this study relied on non-culturing-based metagenomic sequencing, the constructed phylogeny could not fully resolve evolutionary events within these clonal groups. Therefore, a crucial hypothesis regarding the 2023 resurgence remains unresolved based on current evidence: whether it was caused by novel epidemic variants or existing strains acquiring mutations that enhanced transmissibility or immune evasion during the COVID-19 pandemic.

Resistance to macrolides in *M pneumoniae* is primarily caused by specific mutations in the 23S ribosomal RNA (A2063G, A2064G, etc), as confirmed by both in-vitro culturing and clinical isolation.^{10,11} The prevalence of macrolide-resistant *M pneumoniae* varies geographically,

with particularly high rates in Asia (71.22%), compared with relatively lower rates in Europe (2.02%) and the Americas (8.4%).^{8,12} This regional variation underscores the importance of investigating cross-regional transmission dynamics, especially to assess the potential spread of resistant strains to countries with currently low prevalence rates. However, comprehensive genomic analysis has been hampered by the limited availability of public *M pneumoniae* genome data from non-east Asian countries collected after 2015.

Our study presents a large-scale WGS-based investigation of *M pneumoniae* epidemiology in China during the 2023 resurgence. By combining genomic sequences from 436 global strains with 685 newly sequenced strains from our collection, we explored the evolutionary dynamics of this pathogen in recent years. Additionally, by including samples from 11 regions across China, we provided genomic insights into the nationwide transmission patterns of *M pneumoniae* during the 2023 resurgence.

Methods

Study design and participants

We conducted a longitudinal, cross-sectional, genomic study. We collected respiratory specimens, including bronchoalveolar lavage fluid, sputum, and nasopharyngeal swabs, from paediatric patients aged (<18 years) diagnosed with *M pneumoniae* pneumonia from June 24, 2013, to Nov 23, 2023, at Capital Center for Children's Health, Capital Medical University, Capital Institute of Pediatrics (Beijing, China). Additionally, specimens were also obtained from ten other provinces and municipalities (one hospital per region) across China in 2023, from Nov 3, 2023, to Dec 25, 2023 (appendix 1 p 1; appendix 2 p 1). These ten locations covered central, eastern, western, southern, and northern China and were all affected by the resurgence. This study was approved by the research board of the ethics committee of the Capital Institute of Pediatrics (SHERLL2023092). Written informed consent was obtained from the parents or guardians of all patients.

Enrolled participants ranged in age from 5 months to 16 years 1 month at the time of sample collection. The median age of the participants was 6 years 9 months (IQR 4 years 8 months to 8 years 7 months). Of 571 patients with available sample type information, bronchoalveolar lavage fluid constituted the majority of the collected specimen (92.1%), whereas 6.1% of samples were sputum and 1.6% were nasopharyngeal swabs. Of 573 patients with available clinical data for severity determination, 355 were determined to have severe *M pneumoniae* pneumonia. The determination of symptom severity is described in appendix 1 (pp 8–9). Because our sampling strategy only examined paediatric patients and emphasised the collection of bronchoalveolar lavage samples (more frequently obtained in severe cases, 62.9%), we performed additional analyses (appendix 1 pp 8–9) to show that our sampling plan is generalisable in adult patients, as well as across patients with severe and mild *M pneumoniae* pneumonia symptoms. To complement the samples collected in this study, *M pneumoniae* WGS data were downloaded from the National Center for Biotechnology Information (NCBI) Sequence Read Archive (SRA) and GenBank databases (appendix 1 p 1; appendix 2 p 1). To avoid redundancy, if both SRA and assembly data of the same isolate (identified by matching BioSample identification numbers or identical naming from the same laboratory) were available, the assembly data were used for further analysis. Additionally, genetically engineered or passaged strains were also excluded to focus on natural genomic diversity.

Procedures

M pneumoniae was detected using the *Mycoplasma pneumoniae* and Macrolide-Resistant isolates Diagnostic Kit (PCR Fluorescence Probing; Mole BioScience, Jiangsu, China), in accordance with the manufacturer's instructions. Each *M pneumoniae*-positive specimen was cultured in 2 mL *Mycoplasma* selective liquid PPLO broth (Difco, Detroit, MI, USA) and was incubated aerobically at 37°C.

In case of medium colour changing from red to orange, indicative of glucose metabolism, 0.2 mL *M pneumoniae* liquid was transferred to another 2 mL media for subculture.

The total nucleic acid of the isolates was extracted from a 10 mL medium using the QIAamp DNA Mini Kit (Qiagen, Hilden, Germany), in accordance with the manufacturer's instructions. DNA was immediately used or stored at –80°C. The reference strains *M pneumoniae* M129 (ATCC 29342) and FH (ATCC 15531) were used as positive controls, and distilled water was used as a negative control. The extracted DNA of each isolate was subjected to next-generation sequencing, which was performed on the Illumina NextSeq2000 platform, sequencing both ends of a 150 bp-read library. Strains sharing the same metadata (eg, isolation year and patient identification number) were examined and de-duplicated, retaining only one representative.

Genome assembly was performed with SPAdes (version 3.13.0)¹³ in careful mode with a k-mer size of 127 for short-read data from both this study and the NCBI SRA database. In each assembly, contigs shorter than 150 bp were removed. The resulting draft assemblies and publicly available assemblies were further assessed for assembly quality based on genome completeness, and average nucleotide identity against the reference genome. Additionally, ribosomal multilocus sequence typing (MLST),¹⁴ in-silico multilocus variable-number tandem-repeat analysis (MLVA), and in-silico MLST were conducted to confirm species information and detect potential admixture of multiple *M pneumoniae* strains (appendix 1 p 9). 685 newly sequenced genomes and 436 public genomes were retained for downstream analyses (appendix 1 p 1; appendix 2 p 1).

Verified, high-quality *M pneumoniae* sequences were aligned to the reference sequence *M pneumoniae* strain M129 (accession number GCA_000027345) using Snippy, version 4.6.0 for multisequence alignment and SNP calling. Recombination was removed using Gubbins (version 3.3.1).¹⁵ A maximum likelihood tree was constructed using IQ-TREE (version 2.2.5),¹⁶ with the GTR+Gamma model and 1000 ultrafast bootstraps. Interactive trees were generated using Microreact,¹⁷ by integrating the tree file with the metadata.

Statistical analysis

A series of genome-wide association studies (GWAS) were conducted using pyseer (version 1.3.11)¹⁸ using the mixed model to adjust for population structure, and odds ratios and F_{ST} were calculated using R (appendix 1 pp 9–11). Dated trees were constructed by BEAST version 2.6.3,¹⁹ using the Bayesian Skyline model (dimensions 8). The parameters assumed a relaxed log clock model and GTR substitution model, with 300 000 000 Markov Chain Monte Carlo iterations per run (10% burn-in). Effective population size variation plots (Skyline plots) were generated by splitting the dated trees into clades using the splitTree function from the phytools R package (version 2.2.7),²⁰ and applying

See Online for appendix 1

See Online for appendix 2

For more on Snippy see <https://github.com/tseemann/snippy>

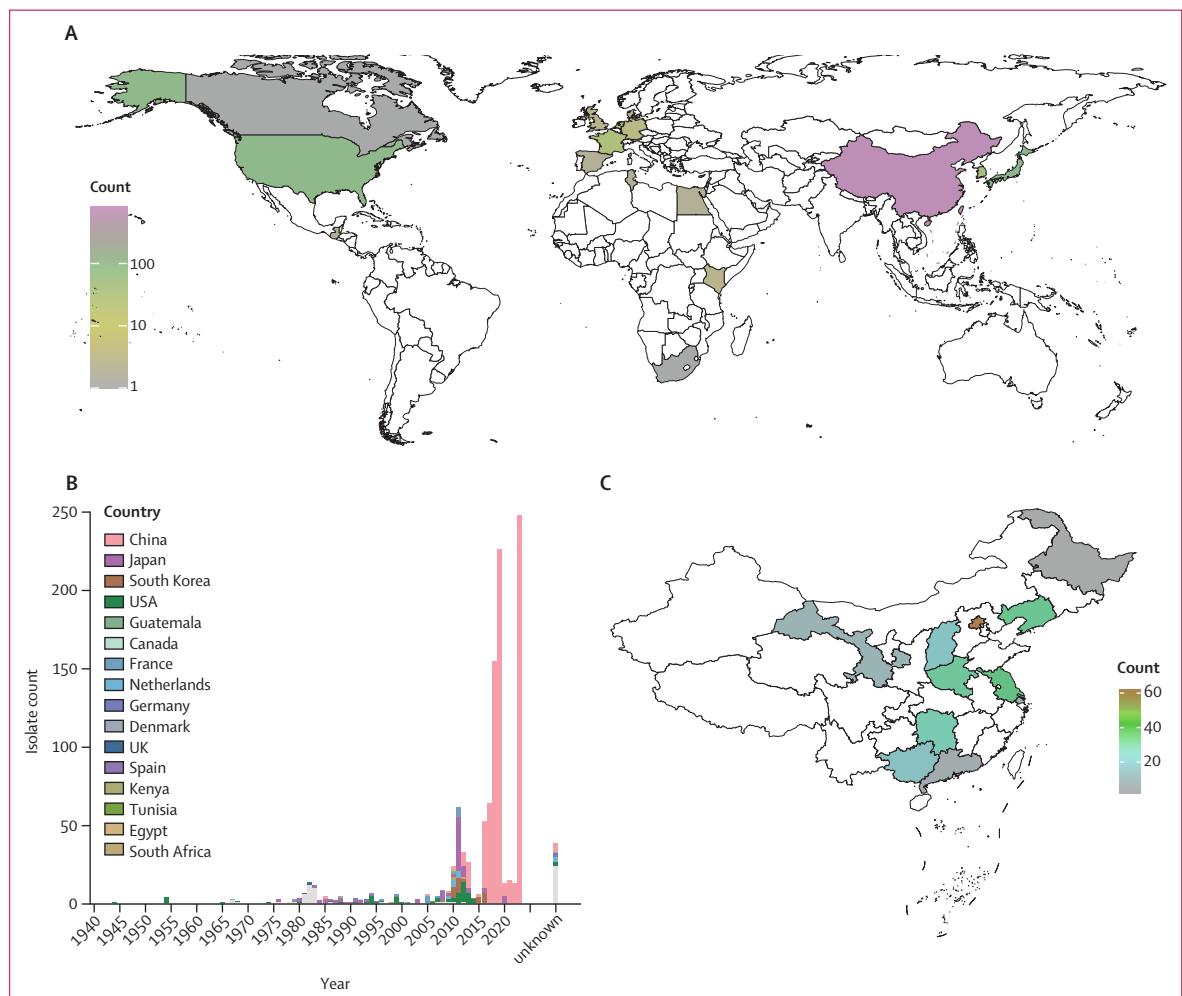


Figure 1: Geographical and temporal distribution of included isolates

(A) Global distribution of all 1121 isolates. (B) Temporal distribution of all isolates. (C) Geographical distribution of 248 isolates collected in China in 2023.

the skygrowth.map function from the skygrowth R package (version 0.3.1).²¹ Geographical distribution was assessed in R (version 4.3.2). The temporal and spatial distribution of clades were visualised using the ggplot2 R package. The ancestral states and migration rate of 2023 T1-2-EC1 isolates in China were estimated using the R package ape (version 5.8)²² using the ace function. The ancestral state likelihoods in internal nodes were visualised using phytools. Additionally, the Kruskal–Wallis test and Fisher’s exact test were performed using R (appendix 1 p 11).

Role of the funding source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

685 *M. pneumoniae* strains were isolated and sequenced from clinical samples collected in mainland China,

comprising 437 isolates from Beijing (2013–22) and 248 isolates from 11 regions in 2023 (Beijing n=62; other provinces n=186). All Beijing isolates were collected at Capital Center for Children’s Health, Capital Medical University, Capital Institute of Pediatrics. The 2023 isolates were collected from June 26 and Dec 25, 2023. Combined with 436 publicly available sequences from global sources spanning four continents and from 1944 to 2020, our complete dataset contained 1121 strains (figure 1; appendix 1 p 1).

Core genome SNP analysis of these 1121 strains revealed five distinct clades across two subtypes, consistent with previous classifications,¹⁵ except for six isolates that could not be assigned to any established clades (figure 2). An interactive version of this phylogenetic tree with complete metadata is available online through Microreact.¹⁷ The two epidemic clusters previously identified by Li and colleagues⁹ were in clades T1-2 and T2-2, which we designate as T1-2-EC1 and T2-2-EC2, respectively. The remaining isolates in these clades form their predecessor

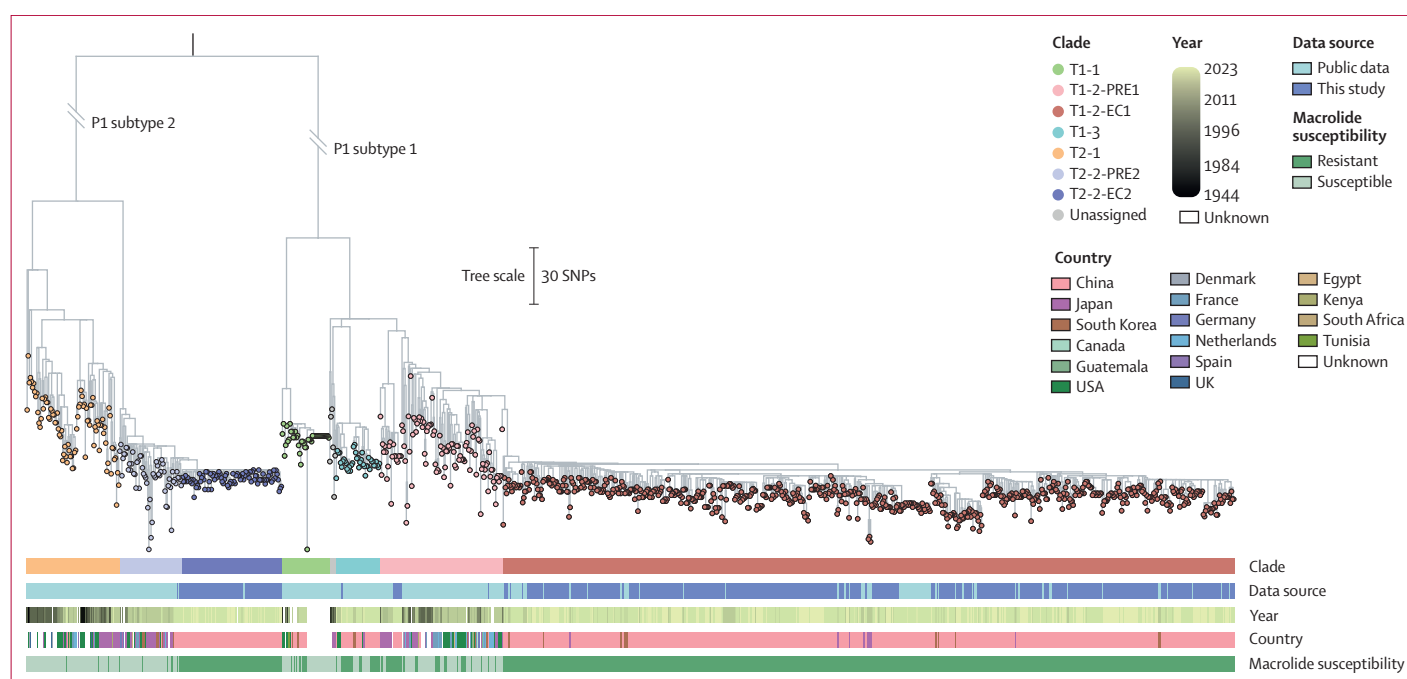


Figure 2: SNP phylogeny of all isolates included in this study

All isolates belong to either the P1 subtype 1 or P1 subtype 2 group. The branches above each group were shortened for visualisation reasons, and the scale bar represents a branch length of 30 SNP differences. The nodes were coloured based on their assigned clade, referencing the definitions provided by Hsieh and colleagues⁴ and Li and colleagues.⁹ The clade information is also recorded in the annotation bands below the tree for direct comparison. Other information is also recorded by the annotation bands including the data source (whether sequenced by this study or retrieved public data), region and continent of sampling, year of sampling, as well as macrolide susceptibility (inferred by detecting known mutations that confer macrolide resistance in *Mycoplasma pneumoniae*). The unassigned clade includes six isolates that could not be assigned to any established clades; these are highlighted as grey nodes and marked by a grey clade bar. SNP=single nucleotide polymorphism.

populations, which we refer to as T1-2-PRE1 and T2-2-PRE2, respectively. Notably, T1-2-EC1 and T2-2-EC2 acquired 17 and four clade-specific point mutations, respectively, that were absent in their predecessor populations (appendix 2 p 2).

Geographical analysis of the phylogenetic distribution revealed that both epidemic clusters (T1-2-EC1 and T2-2-EC2) were exclusively found in east Asia, whereas their predecessor populations and other clades showed wider global distribution (appendix 1 p 2). The earlier clades (T1-1, T1-2-PRE1, T2-1, and T2-2-PRE2) included samples dating back to 1944–91, whereas the epidemic clusters were not collected until recent years—T1-2-EC1 in 2005 and T2-2-EC2 in 2016. We found T2-2-EC2 isolates from 2016 to 2018, indicating that this cluster emerged in China at least 3 years earlier than previously reported.^{4,9}

All isolates in both T1-2-EC1 and T2-2-EC2 carried the A2063G mutation in the 23S ribosomal RNA, a mutation associated with macrolide resistance corresponding to the results of Li and colleagues⁹ (figure 2). This mutation was also frequent in clade T1-3 (63% of isolates), moderately present in T1-1 (23%) and T1-2-PRE1 (32%), and less common in T2-1 (5%) and T2-2-PRE2 (12%).

Examination of temporal evolution patterns revealed distinct succession dynamics for the two epidemic clusters (figure 3; appendix 1 p 3). Before 2010, T1-2-PRE1 was the predominant T1-2 population with global distribution. The

first sequenced T1-2-EC1 strain (*M pneumoniae* strain M29, GCA_000733995) was isolated in Beijing in 2005, with subsequent isolates identified from 2010 onward across east Asia. From 2010 to 2015, T1-2-EC1 coexisted with T1-2-PRE1 in east Asia, but by 2016–19, T1-2-EC1 had become predominant while T1-2-PRE1 declined. After 2020, T1-2-PRE1 was undetected, leaving T1-2-EC1 as the sole T1-2 population in east Asia. T2-2-EC2 showed a similar, but later emergence pattern (figure 3). T2-2-PRE2 was the only T2-2 population detected before 2015. T2-2-EC2 first appeared in 2016 and rapidly increased in frequency, becoming predominant by 2018 and the sole T2-2 population after 2020.

These population dynamics were further supported by the Bayesian Skyline analysis (figure 4). The effective population size of T1-2-PRE1 peaked in the 1980s and maintained this level until approximately 2005 before declining. The most recent common ancestor of T1-2-EC1 was estimated to have existed around 1997, with the population experiencing rapid growth during 2010–16 before beginning to decline. For T2-2-PRE2, the effective population size peaked around 2010 before declining, while T2-2-EC2 emerged around 2014 and showed rapid growth until recently, when its growth rate decreased.

To identify potential genetic factors associated with the 2023 resurgence, we investigate genomic variations specific to the 2023 isolates within each epidemic cluster through a

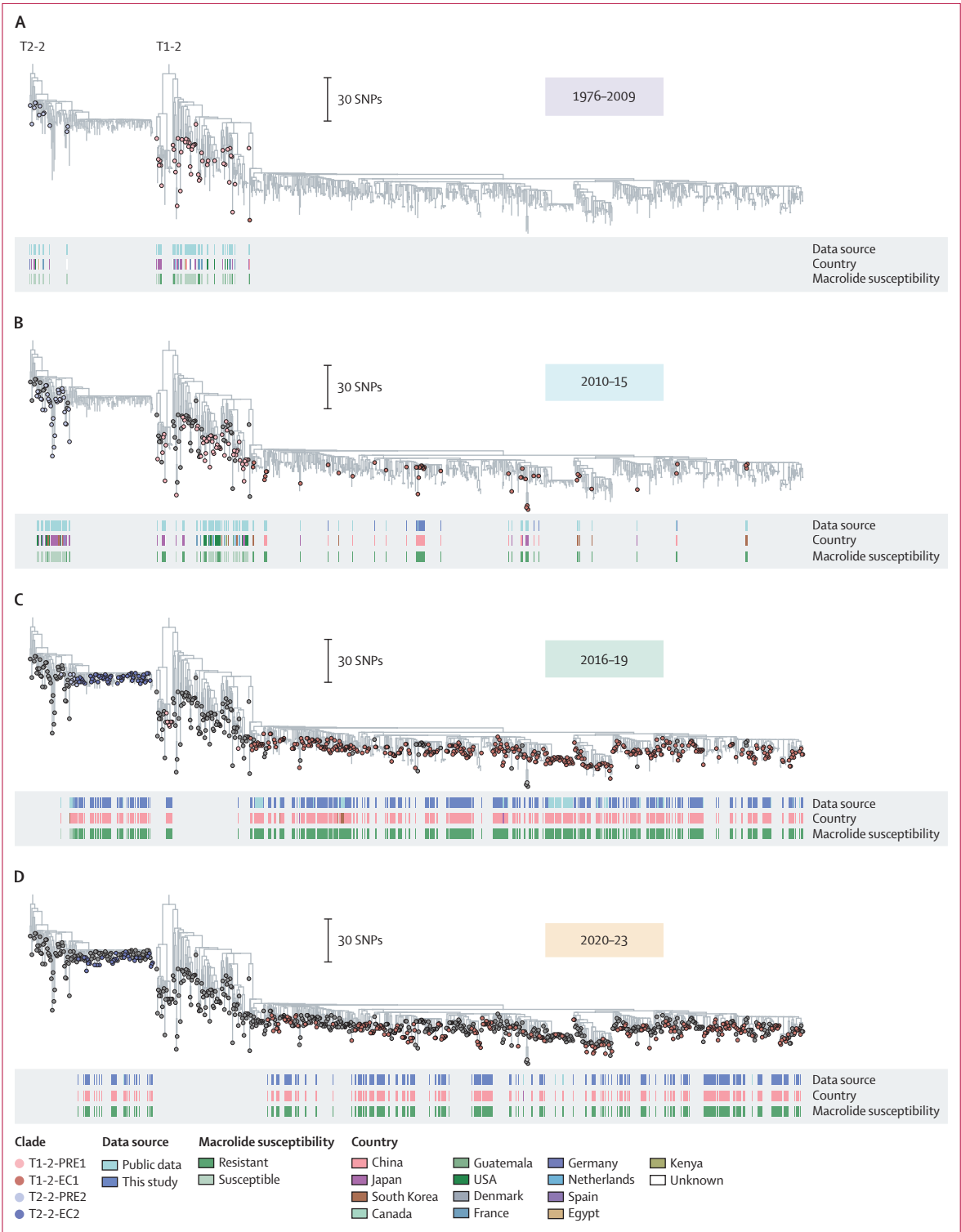


Figure 3: The time-separated subtree panels reveal the evolution history of T1-2-EC1 and T2-2-EC2
Panels (A–D) depict the same subtrees of T1-2 and T2-2, each corresponding to a different time period (1976–2009, 2010–15, 2016–19, and 2020–23). In each panel, nodes that were isolated in the corresponding period are coloured and annotated accordingly. Nodes that were isolated before the period are coloured grey and not annotated. The scale bar indicates a branch length of 30 SNP differences. EC=epidemic cluster. SNP=single nucleotide polymorphism.

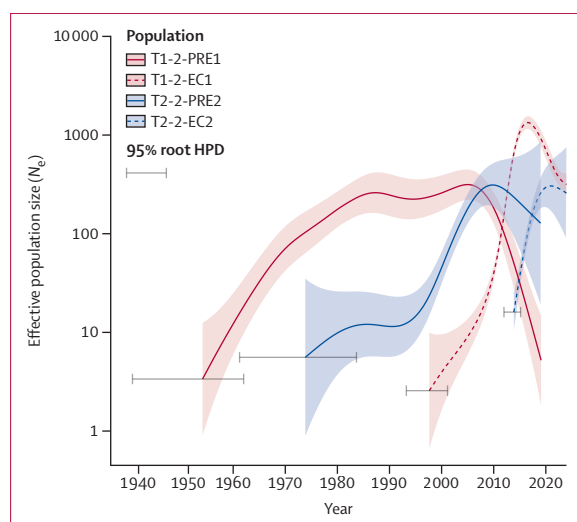


Figure 4: Variation of effective population sizes in both predecessors and both epidemic clusters, inferred from time-based phylogenies
Each line represents one population. Colour separates T1-2 (red) and T2-2 (blue), while line type distinguishes predecessors (full) and epidemic clusters (dashed). Each line starts at the estimated emergence time of the population and ends at the timepoint of the latest isolation. The 95% CIs of the effective population sizes are indicated by transparent bands. The error bars represent the 95% HPDs of clade root dates. HPD=highest probability density.

series of GWAS. Our analysis revealed no unitigs with complete or nearly complete association with the 2023 isolates in either T1-2-EC1 or T2-2-EC2 (appendix 2 pp 3–4). Time-calibrated phylogenies further supported this finding, showing no evidence of variants with selective advantages emerging within either epidemic cluster (appendix 1 pp 4–5). These results indicate that the 2023 resurgence of *M pneumoniae* in China represents a continuation of the pre-COVID-19 epidemic, rather than the emergence of novel variants.

To examine the nationwide migration of *M pneumoniae*, we collected isolates (2023 only) from 11 provinces and municipalities across China (figure 1). These isolates were either identified as T1-2-EC1 (224 isolates from ten regions), T2-2-EC2 (23 isolates from seven regions), or T1-3 (one isolate), with T2-2-EC2 appearing more common in northern China (appendix 1 p 6). We focused further on T1-2-EC1 to investigate nationwide transmission patterns due to its broader distribution and larger number of isolates. A maximum likelihood SNP tree constructed from 2023 T1-2-EC1 isolates showed rapid inter-regional mixing within China (figure 5). The estimated migration rate was 3.169 transmission events per year between any two regions (SE 1.941), translating to a total migration rate of 28.521 events per year across all regions. Ancestral state reconstruction showed highly uniform location probabilities for internal nodes predating summer 2023 (~10% for each of the ten regions), with only the most recent nodes showing evidence of local clustering (appendix 1 p 7). This pattern indicates rapid decay of geographical signals over time due to extensive mixing between regions (figure 5).

Despite this rapid mixing, analysis of pairwise SNP distances between isolates revealed subtle geographical structuring. While the overall distribution of pairwise SNP distances did not differ significantly between same-region and different-region isolate pairs (Kruskal–Wallis test; $p=0.78$), closely related pairs (≤ 7 SNPs) were enriched within the same geographical regions (Fisher's exact test; $p<10^{-6}$; figure 5; appendix 2 p 5). This suggests that while cross-regional transmission occurs rapidly, recently diverged lineages still show significant geographical association.

Discussion

This study provides, to our knowledge, the most comprehensive high-resolution genomic analysis of the *M pneumoniae* resurgence in China to date, doubling the number of publicly available *M pneumoniae* genomes with the addition of 685 high-quality draft genomes from mainland China (2013–23). Our whole-genome analysis shows that the 2023 resurgence was not caused by newly emerged variants, but rather by the resurgence of two previously established macrolide-resistant *M pneumoniae* epidemic clusters. We characterised the evolutionary dynamics of epidemic clusters, documenting their population replacement patterns and quantifying transmission rates. The data reveal rapid inter-regional mixing of *M pneumoniae* populations across China.

Our high-resolution phylogenetic analysis provides new insights into the evolutionary history of these epidemic clusters. T1-2-EC1, previously identified as a product of clonal expansion of a macrolide-resistant T1-2 strain,^{4,9} emerged earlier than previously thought, with our dated tree suggesting an origin in the 1990s despite the first isolate being collected in 2005. Similarly, we show that T2-2-EC2 emerged around 2014, with the earliest isolate in our collection dating to 2016, predating the first reported isolation in 2019.^{4,9} This finding confirms the hypothesis that this variant had been spreading throughout China before 2020 and aligns with a previous surveillance study in Beijing that found a substantial increase in macrolide resistance of MLVA type 3-5-6-2 isolates (mostly corresponding to T2-2; appendix 2 p 1) from 60% in 2016 to 93.5% in 2019.²³

Both T1-2 and T2-2 clades underwent a complete replacement of their predecessor populations in mainland China, with neither T1-2-PRE1 nor T2-2-PRE2 isolates detected after 2020. These population shift events are further supported by our Bayesian Skyline analysis, which shows declining effective population sizes for predecessor clusters followed by growth of their respective epidemic clusters. We observed a gradual genotype shifting trend in recent years, with T2-2-EC2 showing a more gradual decline in effective population size compared with the steeper decline of T1-2-EC1. This differential rate of decline might eventually lead to a shift in the relative proportions of these two epidemic clusters within the population. Experimental studies would be required to assess competitive

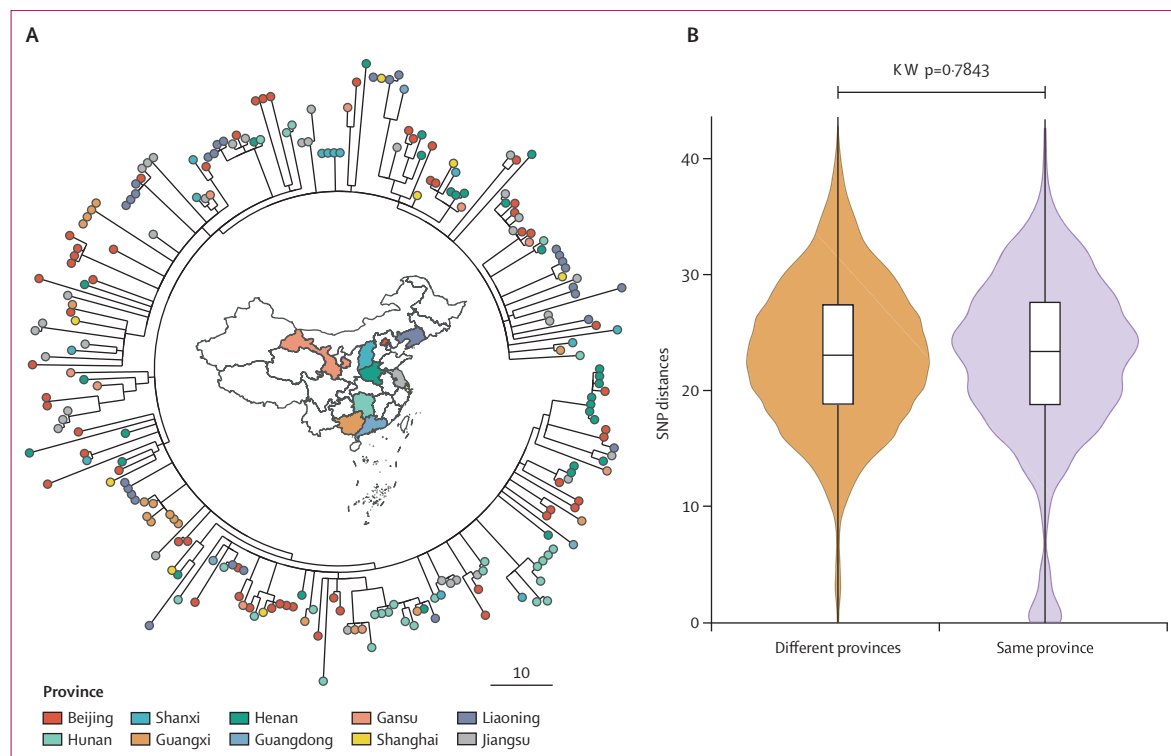


Figure 5: Analysis of the geographical distribution and migration of 2023 isolates

(A) SNP phylogeny of all T1-2-EC1 isolates collected in 2023. Nodes were coloured by the province and municipality in which they were isolated. (B) Violin plot of pairwise SNP distance distributions in the 2023 phylogeny, separated by same-location and different-location isolate pairs. The median and quantiles in each group are marked accordingly. EC=epidemic cluster. SNP=single nucleotide polymorphism.

dynamics between these two epidemic clusters and predict future trends.

The timing of these population shifts might be linked to changes in clinical practice. Due to its significantly higher efficacy and longer half-life,²⁴ azithromycin replaced erythromycin and became the first-line treatment recommendation in the clinical guidelines for treating paediatric community-acquired pneumonia patients (2007) in mainland China.^{25,26} This timing coincides approximately with the decline of T1-2-PRE1 and precedes the rise of T1-2-EC1. We hypothesise that this shift in clinical prescription might have substantially increased selective pressure, triggering the decline of predecessor populations and subsequent dominance of the epidemic clusters. This relationship between antibiotic use and bacterial population dynamics might explain why T1-2-EC1, although emerging in the 1990s, did not become prevalent until the early 2010s, whereas T2-2-EC2 emerged in the 2010s and rapidly became dominant.

Both epidemic clusters exhibit convergent evolution by acquisition of the A2063G macrolide resistance mutation at the 23S ribosomal RNA region. However, this mutation alone cannot fully explain their evolutionary success, as resistance strains also exist within predecessor populations. We identified 17 clade-specific mutations unique to T1-2-EC1 and four specific to T2-2-EC2 (appendix 2 p 2),

which might contribute to their competitive advantage through mechanisms such as enhanced transmission, metabolic adaptation, or compensating for the fitness cost associated with macrolide resistance. Such fitness costs have been observed in *M hyopneumoniae*, a related species that infects pigs.²⁷ Although functional validation of these mutations is needed, they are promising targets for future studies to investigate the mechanisms causing these epidemic clusters to outcompete their predecessors.

Our GWAS and dated phylogeny analysis show that the strains responsible for the 2023 resurgence in China did not acquire new genomic mutations contributing to their re-emergence. The resurgence appears to be a continuation of the pre-COVID-19 epidemic rather than a result of newly emerged strains, which aligns with reports from China and worldwide surveillance indicating no significant changes in disease severity or mortality rates associated with *M pneumoniae* infection before and after the pandemic.⁸ This finding suggests non-genomic causes to explain the observed epidemic patterns. Therefore, the 2023 resurgence likely results from a complex interplay of factors, including the waning herd immunity during the pandemic, often termed immunity debt,²⁸ combined with the pathogen's natural long generation time, and human social behaviour, such as changes in health-care-seeking attitude and the resumption of normal population mobility

patterns. This complexity highlights the need for interdisciplinary approaches combining genomic, immunological, and epidemiological methods. Future models incorporating more comprehensive surveillance data that include both human immunity status and pathogen genetic diversity would help to investigate annual fluctuation patterns and trends of *M pneumoniae*.

Although T1-3 was reported to be highly prevalent in Taiwan,⁴ only two isolates from mainland China were sequenced in this study (2019 Beijing and 2023 Guangdong). Given the high frequencies of macrolide-resistant mutations in T1-3, this clade might potentially develop into a third macrolide-resistant epidemic cluster. Our study estimated a high rate of local migration within China, resulting in rapid mixing between provincial populations and weak geographical structure. However, from a global perspective, the significantly lower macrolide resistance rates in Europe and the Americas suggest that T1-2-EC1 and T2-2-EC2 might not have spread to these regions.¹² A study from Denmark reported only 1.5% macrolide resistance among 2023 clinical isolates, supporting this hypothesis.²⁹ Although migration within China is rapid, intercontinental transmission is substantially slower, explaining why these epidemic clusters remain predominantly limited to east Asia. To confirm this hypothesis, more clinical *M pneumoniae* strains need to be sequenced in non-Asian countries. Furthermore, as long as macrolide continues to be used to treat paediatric community-acquired pneumonia, it is reasonable to forecast that the epidemic clusters will eventually spread in these countries, causing macrolide resistance to become a global issue rather than a regional one. A 2018 report of 90% resistance in MLVA type 4-5-7-2 (which corresponds to either T1-1 or T1-2; appendix 2 p 1) strains from Iran might indicate that T1-2-EC1 has already reached west Asia.³⁰

Our study has limitations. First, our sequenced strains are collected exclusively from clinical samples of paediatric patients and emphasised on those with severe *M pneumoniae* pneumonia. Although we showed that our findings are most likely to be generalisable in other patients, we acknowledge that incorporating samples of adult patients, community surveillance, and asymptomatic carriers would provide a more comprehensive representation of *M pneumoniae* population structure. Second, our sampling was limited to mainland China. Including new samples from Europe, the Americas, and other regions would substantially enhance our understanding of the global resurgence of this pathogen in 2023, including the geographical distribution of each genomic cluster and cross-continental transmission dynamics. Finally, although we identified clade-specific mutations, functional studies are needed to confirm their roles in macrolide resistance and fitness.

In summary, this study vastly improved the phylogenetic resolution for the two epidemic clusters by providing the genomes of 579 (85.3%) of 679 T1-2-EC1 and 91 (97.8%) of 93 T2-2-EC2 isolates (appendix 2 p 1). We show that the

M pneumoniae epidemic in China was a process in which epidemic clusters outcompeted their predecessor populations, which led to the 2023 resurgence without evolving into novel variants. We highlight the need for more isolates to be collected and sequenced from non-Asian countries to better understand international transmission dynamics of this pathogen, calling for increased monitoring efforts in the future using WGS approaches.

Contributors

YCu and JY conceived and designed the study. CY, YCh, XJ, XR, CH, ZZhu, SL, HZha, YWa, LiZho, DC, ZH, YY, LH, and LC collected the *Mycoplasma pneumoniae* samples. CY, BD, HZha, YF, JC, and YX were responsible for strain revival and cultivation. CY, YCh, ZW, YS, LZha, YK, and GX performed DNA extraction. YG, MC, and GP constructed the DNA library and did the DNA sequencing. ST, YWu, XD, ZZha, HZho, and KM conducted bioinformatic analyses. ST, YCu, and XD wrote the manuscript. YCu and JY supervised the project. All authors had full access to all the data in the study and accept responsibility for the decision to submit for publication. YCu, JY, CY, ST, and YWu accessed and verified the underlying data.

Declaration of interests

We declare no competing interests.

Data sharing

The genomic sequence data from this study have been submitted to the China National Microbiology Data Center under the BioProject accession number NMDC10019358. The code relevant to this manuscript is publicly available on GitHub (github.com/davidtong28/misc_scripts).

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